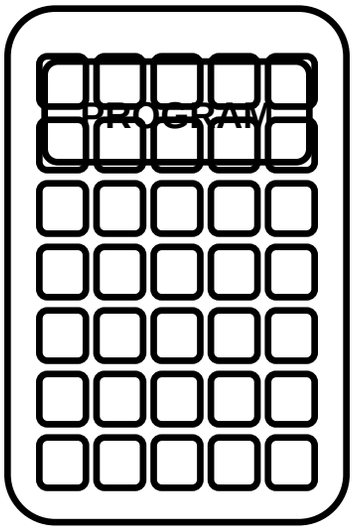
CASIO fx-50FH II



S5 PROGRAM LIBRARY

中五計算機程式 · 精選版

22 tested programs · 10 memory-safe packs · 680-byte aware

Program 01 一元二次方程工具
Quadratic X-Ray

Program 02 等差及等比數列
Sequence Engine

Program 03 坐標幾何及圓
Coordinate Lab

Program 04 三角方程解
Trig 360

Program 05 聯立二元一次方程
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Program 06 三點定圓
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Program 20 二項及泊松分佈
Binomial and Poisson

Program 21 三維向量
3D Vector Toolkit

Program 22 正態分佈概率
Normal Probability

Built from the canonical tested web catalog · Revised 26 September 2026

PRGM Command Reference

程式模式按鍵速查

Command	Physical keys	Command	Physical keys
?	SHIFT 3 1	ClrMemory	SHIFT 9 1
: / new line	SHIFT 3 3 or EXE	ClrStat	SHIFT 9 1
◢ output	SHIFT 3 4	Goto	SHIFT 3 ▶ ▶ ▶ 1
→ assign	SHIFT 3 2	Lbl	SHIFT 3 ▶ ▶ ▶ 2
⇒	SHIFT 3 ▶ 1	If	SHIFT 3 ◀ 1
=	SHIFT 3 ▶ 2	Then	SHIFT 3 ◀ 2
≠	SHIFT 3 ▶ 3	Else	SHIFT 3 ◀ ◀ 1
> / <	SHIFT 3 ▶ ▶ 1 / 2	IfEnd	SHIFT 3 ◀ ◀ 1
≥ / ≤	SHIFT 3 ▶ ▶ 3 / 4	While	SHIFT 3 ▶ ▶ ▶ 1
A B C D	ALPHA + letter key	WhileEnd	SHIFT 3 ▶ ▶ ▶ 2
X Y M	ALPHA +) , M+	For / To / Step	SHIFT 3 ◀ ◀ ◀ 1/2/3
Rnd(	SHIFT 0	Next	SHIFT 3 ◀ ◀ ◀ ◀ 1
Fix / Sci / Norm	SHIFT MODE ▶ 1/2/3	Abs(	SHIFT)
Pol(/ Rec(	SHIFT + / SHIFT -	DT	M+ in SD/REG

Tip: an ordinary minus sign is accepted in Program Studio; use the physical (-) key for negative input on the calculator.

Program 1

Quadratic X-Ray

一元二次方程工具

用法

一次輸入係數，即時計算判別式、頂點及所有實根。

INPUT A=a，B=b，C=c

MODE COMP · 61 bytes

OUTPUT 判別式、頂點 (h,k)、實根

Program steps

```
?→A
?→B
?→C
B²-4AC→D◢-B÷2÷A→X◢-D÷4÷A◢If D≥0
Then X+√(D)÷2÷A◢2X-Ans
```

示範 Example

```
TYPE  A=1, B=-5, C=6
GET   Δ=1, h=2.5, k=-0.25, roots 3 and 2
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 2

Sequence Engine

等差及等比數列

用法

處理等差、有限等比及無限等比數列的項及和。

INPUT M=類型；A=首項；B=公差/公比；C=n

MODE COMP · 85 bytes

OUTPUT 指定項及部分和/無限和

Program steps

```
?→M
?→A
?→B
0→D
M<3⇒?→C
M=1⇒A+(C-1)B→D
M=2⇒AB^(C-1→D
D◢C(A+D)÷2→X
M≠1⇒(A-BD)÷(1-B→X
X
```

示範 Example

```
TYPE  M=1, A=2, B=3, C=5
GET   T=14, S=40
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 3

Coordinate Lab

坐標幾何及圓

用法

計算兩點距離、中點、斜率及截距，或由一般式求圓心及半徑。

INPUT M=1 兩點；M=2 圓的一般式係數

OUTPUT 距離、中點、斜率、截距／圓心、半徑

MODE COMP · 100 bytes

Program steps

```
?→M
?→A
?→B
?→C
If M=2
Then -A÷2◢-B÷2◢√(A²+B²-4C)÷2
Else ?→D
C-A→X
D-B→Y
√(X²+Y²)◢(A+C)÷2◢(B+D)÷2◢Y÷X◢B-AAns
```

示範 Example

```
TYPE  M=1, A=0, B=0, C=3, D=4
GET   distance 5, midpoint (1.5,2), slope 1.333333333, intercept 0
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 4

Trig 360

三角方程解

用法

求 0° 至 360° 內所有 sin、cos 或 tan 方程的解。

INPUT M=1/2/3；D=已分離的三角函數值

MODE COMP · 103 bytes

OUTPUT 區間內所有不同角度

Program steps

```
Deg
?→M
?→D
sin⁻¹(D(M<3→B
tan⁻¹(D(M=3))-90(M=2→C
-1E-99→X
For 0→A To 3
C+180A+B(-1)^(A→Y
If (Y>X)(Y≤360
Then Y→X◢IfEnd
Next
X
```

示範 Example

2sin θ=1 → TYPE M=1, D=0.5 → GET 30°, 150°
cos θ=1 → TYPE M=2, D=1 → GET 0°, 360°
tan θ=-1 → TYPE M=3, D=-1 → GET 135°, 315°

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 5

Simultaneous 2×2 Solver

聯立二元一次方程

用法

輸入兩條直線方程的六個係數，求唯一交點。

INPUT A,B,C 及 D,X,Y 為兩式係數

MODE COMP · 51 bytes

OUTPUT x、y

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
AX-BD→M
(CX-BY)÷M◢(AY-CD)÷M
```

示範 Example

```
TYPE  A=2, B=1, C=7, D=1, X=-1, Y=2
GET   x=3, y=1
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 6

Circle Through Three Points

三點定圓

用法

由三個不共線點求圓心及半徑。

INPUT A,B；C,D；X,Y 為三點坐標

MODE COMP · 114 bytes

OUTPUT 圓心、半徑

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
C-A→C
D-B→D
X-A→X
Y-B→Y
(CX+DY)÷(CY-DX→M
Pol((C+X-M(Y-D))÷2,(D+Y-M(C-X))÷2→M
Rec(M,Y
A+X◢B+Y◢M
```

示範 Example

```
TYPE  A=0, B=0, C=4, D=0, X=0, Y=6
GET   centre (2,3), radius 3.605551275
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 7

Growth and Decay Solver

增長與衰減

用法

在複式增長或衰減模型中求終值、時間或率；期數輸入負數可求初值。

INPUT M=1/2/3；其餘輸入已知數值

MODE COMP · 73 bytes

OUTPUT 所選未知量

Program steps

```
?→M
?→A
?→B
?→C
1+B÷100→Y
M=1⇒AY^(C→X
M=2⇒log(Y,C÷A→X
M=3⇒100(C√(B÷A)-1→X
X
```

示範 Example

```
TYPE  M=1, A=1000, B=5, C=3       → GET 1157.625
TYPE  M=2, A=1000, B=10, C=1210   → GET 2 periods
TYPE  M=3, A=1000, B=1210, C=2    → GET 10%
TYPE  M=1, A=1157.625, B=5, C=-3  → GET start 1000
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 8

Combined Population Statistics

合併統計數據

用法

只有兩組人數、平均數及母體標準差時，合併兩組數據。

INPUT A,B,C 及 D,X,Y 為兩組 n, μ, σ

MODE COMP · 67 bytes

OUTPUT 合併平均數及母體標準差

Program steps

```
?→A
?→B
?→C
?→D
?→X
?→Y
A+D→M
(AB+DX)÷M◢√((A(C²+B²)+D(Y²+X²))÷M-Ans²
```

示範 Example

```
TYPE  A=2, B=10, C=2, D=2, X=14, Y=2
GET   combined mean 12, combined SD 2.828427125
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 9

Line–Circle Intersections

直線與圓的參數 k 交點

用法

當 k 出現在圓或直線的係數時，代入指定 k 值並求數值交點。

INPUT M=k；圓 A,B,C；直線 D,X,Y；係數可含 M

MODE COMP · 124 bytes

OUTPUT 一或兩組 (x,y)；切線只輸出一組

Program steps

```
?→M
?→A
?→B
?→C
?→D
?→X
?→Y
-A÷2→A
-B÷2→B
A²+B²-C→C
D²+X²→M
(AD+BX+Y)÷M→Y
A-YD→A
B-YX→B
√(C÷M-Y²→C
A-CX◢B+CD◢If C
Then A+CX◢B-CD
```

示範 Example

```
TYPE  M=1, A=M, B=1, C=-4, D=1, X=M, Y=0
GET   (-√2,√2), (√2,-√2)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 10

Sequence Target Solver

數列目標項數

用法

求部分和首次達到指定目標時的最小正整數 n。

INPUT M=AP/GP；A=首項；B=公差/公比；C=目標

MODE COMP · 62 bytes

OUTPUT 最小 n 及對應 S

Program steps

```
ClrMemory
?→M
?→A
?→B
?→C
While Y<C
X+1→X
Y+A→Y
M=1⇒A+B→A
M=2⇒AB→A
WhileEnd
X◢Y
```

示範 Example

```
TYPE  M=1, A=3, B=2, C=100
GET   n=10, S₁=120
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 11

Complete Cubic Solver

一元三次方程完整求根

用法

毋須先知道因式，直接求一元三次方程的所有實根。

INPUT A,B,C,M 為 $Ax^3+Bx^2+Cx+D=0$ 的係數

MODE COMP · 134 bytes

OUTPUT 一個或三個實根（重根會重複顯示）

Program steps

```
Deg
?→A
?→B
?→C
?→M
B÷3÷A→B
B²-C÷3÷A→X
B³-(BC-M)÷2÷A→D
D²-X³→M
If M≤0
Then Pol(-D,√(-M
For 0→M To 2
2(X)cos(Y÷3-120M)-B◢Next
Else √(M→M
(M-D)-(D+M)-B
```

示範 Example

$x^3+6x^2+11x+6=0$
TYPE A=1, B=6, C=11, M=6
GET -1, -2, -3

$2x^3+7x^2+5x+1=0$
TYPE A=2, B=7, C=5, M=1
GET approximately -0.381966, -0.5, -2.618034

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 12

Without-Replacement Probability

不放回抽樣概率

用法

一次過以組合公式計算恰好、至少及至多指定數目的概率。

INPUT A,B=兩類數量；C=抽取數；D=目標

MODE COMP · 86 bytes

OUTPUT P(=D)、P(≥D)、P(≤D)

Program steps

```
?→A
?→B
?→C
?→D
(A+B)nCr C→Y
0→M
For (C-B)(C>B→X To D
A nCr X B nCr(C-X)÷Y M+
Next
A nCr D B nCr(C-D)÷Y◢1-M+Ans◢M
```

示範 Example

```
TYPE  A=5, B=4, C=2, D=2  → GET 0.2777777778, 0.2777777778, 1
TYPE  A=5, B=4, C=2, D=1  → GET 0.5555555556, 0.8333333333, 0.7222222222
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 13

Transformed-Roots Builder

變換根方程

用法

不先求原根，直接建立線性變換根或平方根的新二次方程。

INPUT M=模式；A,B,C=原式；模式 1 再輸入 X,Y

MODE COMP · 73 bytes

OUTPUT 新首一二次方程的 P、Q

Program steps

```
?→M
?→A
?→B
?→C
If M=1
Then ?→X
?→Y
XB÷A-2Y◢X²C÷A-YAns-Y²
Else C÷A→X
2X-(B÷A)²◢X²
```

示範 Example

```
TYPE  M=1, A=1, B=-5, C=6, X=2, Y=1
GET   P=-12, Q=35
WRITE z²-12z+35=0
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 14

Tangents from a Point to a Circle

圓外一點的切線

用法

由圓心、半徑及圓外點求切線長及兩個切點。

INPUT A,B=圓心；C=半徑；D,X=圓外點

MODE COMP · 97 bytes

OUTPUT 切線長及兩個切點

Program steps

```
?→A
?→B
?→C
?→D
?→X
D-A→D
X-B→X
D²+X²→M
√(M-C²→Y◢C÷M→M
A+M(CD-XY)◢B+M(CX+DY)◢A+M(CD+XY)◢B+M(CX-DY
```

示範 Example

```
TYPE  A=0, B=0, C=5, D=13, X=0
GET   tangent length 12
      contact points (1.923076923,4.615384615)
                     (1.923076923,-4.615384615)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 15

Annuity and Loan Engine

年金與貸款

用法

先求貸款每期供款，再求期末供款後的儲蓄終值或尚欠餘額。

INPUT A=本金；B=每期利率%；C=期數；D=供款（還款輸入負數）

MODE COMP · 53 bytes

OUTPUT 每期供款，然後結餘

Program steps

```
?→A
?→B
?→C
B%→B
(1+B)^(C→X
ABX÷(X-1)◢?→D
AX+D(X-1)÷B
```

示範 Example

```
TYPE  A=1000, B=1, C=12   → GET 88.84878868 (payment, ignore)
TYPE  D=100               → GET future value 2395.075331
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 16

Four Centres of Any Triangle

任意三角形四心

用法

輸入任意三個不共線頂點，求形心、內心、垂心及外心。

INPUT REG 模式；A,B；C,D；X,Y 為三點坐標

MODE REG · 217 bytes

OUTPUT G、I、H、O 四組坐標

Program steps

```
ClrStat
?→A
?→B
A,B DT
?→C
?→D
?→X
?→Y
C-A→C
D-B→D
X-A→X
Y-B→Y
Σx+(C+X)÷3◢Σy+(D+Y)÷3◢√(X²+Y²→A
√(C²+D²→B
A+B+√((C-X)²+(D-Y)²→M
Σx+(AC+BX)÷M◢Σy+(AD+BY)÷M◢(CX+DY)÷(CY-DX→M
Σx+M(Y-D◢Σy+M(C-X◢Σx+(C+X-M(Y-D))÷2◢Σy+(D+Y-M(C-X))÷2
```

示範 Example

```
TYPE  A=0, B=0, C=24, D=0, X=12, Y=9
GET   G=(12,3), I=(12,4), H=(12,16), O=(12,-3.5)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 17

Triangle Solver

解三角形

用法

以餘弦定理由三邊或兩邊及夾角求第三邊、三個角及面積。

INPUT M=1 三邊 a,b,c；M=2 a,b 及夾角 C

MODE COMP · 99 bytes

OUTPUT c、角 A、B、C、面積

Program steps

```
Deg
?→M
?→A
?→B
?→C
M=2⇒√(A²+B²-2ABcos(C→C
C◢cos⁻¹((B²+C²-A²)÷(2BC→X◢cos⁻¹((A²+C²-B²)÷(2AC→Y◢180-X-Y◢ABsin(Ans)÷2
```

示範 Example

```
3-4-5 triangle     TYPE M=1, A=3, B=4, C=5
                   GET  c=5, A=36.86989765°, B=53.13010235°, C=90°, area 6
a=5, b=7, C=60°    TYPE M=2, A=5, B=7, C=60
                   GET  c=6.244997998, A=43.89788625°, B=76.10211375°, C=60°,
                        area 15.15544457
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 18

Point and Line

點與直線

用法

求點到直線 Ax+By+C=0 的垂直距離、垂足及對稱點。

INPUT A,B,C=直線係數；X,Y=點

MODE COMP · 72 bytes

OUTPUT 距離、垂足、對稱點

Program steps

```
?→A
?→B
?→C
?→X
?→Y
A²+B²→M
(AX+BY+C)÷M→D
Abs(D)√(M)◢X-AD◢Y-BD◢X-2AD◢Y-2BD
```

示範 Example

```
TYPE  A=3, B=4, C=-25, X=0, Y=0
GET   distance 5, foot (3,4), image (6,8)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 19

Polygon Area

多邊形面積

用法

以鞋帶公式由頂點坐標求任意多邊形的面積。

INPUT 依次輸入各頂點，最後再輸入第一個頂點

MODE COMP · 69 bytes

OUTPUT 面積

Program steps

```
ClrMemory
?→A
?→B
A→X
B→Y
Lbl 1
?→C
?→D
XD-YC M+
C→X
D→Y
(X≠A)+(Y≠B)⇒Goto 1
Abs(M)÷2
```

示範 Example

```
TYPE  A=0, B=0, then C,D = 4,0 · 4,3 · 0,3 · 0,0
GET   area 12
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 20

Binomial and Poisson

二項及泊松分佈

用法

（M1）計算二項或泊松分佈的恰好、至少及至多概率。

INPUT Y=1 二項 (A=n, B=p)；Y=2 泊松 (A=λ)；C=r

MODE COMP · 93 bytes

OUTPUT P(=r)、P(≥r)、P(≤r)

Program steps

```
ClrMemory
?→Y
?→A
Y=1⇒?→B
?→C
For 0→X To C
Y=1⇒A nCr X B^(X)(1-B)^(A-X→D
Y=2⇒e^(-A)A^(X)÷X!→D
D M+
Next
D◢1-M+D◢M
```

示範 Example

```
B(10, 0.3), r=3   TYPE Y=1, A=10, B=0.3, C=3
                  GET  0.266827932, 0.6172172136, 0.6496107184
Po(2), r=1        TYPE Y=2, A=2, C=1
                  GET  0.2706705665, 0.8646647168, 0.4060058497
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 21

3D Vector Toolkit

三維向量

用法

（M2）計算數量積、夾角、向量積及平行四邊形面積。

INPUT A,B,C=u；D,X,Y=v

MODE COMP · 91 bytes

OUTPUT u·v、夾角、|u×v|、u×v

Program steps

```
Deg
?→A
?→B
?→C
?→D
?→X
?→Y
√((A²+B²+C²)(D²+X²+Y²→M
AD+BX+CY◢cos¹(Ans÷M◢Msin(Ans)◢BY-CX◢CD-AY◢AX-BD
```

示範 Example

```
TYPE  A=1, B=2, C=3, D=4, X=5, Y=6
GET   u·v=32, angle 12.93315449°, |u×v|=7.348469228, u×v=(-3,6,-3)
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.

Program 22

Normal Probability

正態分佈概率

用法

毋須查表，計算標準分及正態分佈兩側概率。

INPUT A=μ；B=σ；C=x

MODE COMP · 83 bytes

OUTPUT z、P(X<x)、P(X>x)

Program steps

```
?→A
?→B
?→C
(C-A)÷B→X◢X→Y
X→D
1→C
While Y
C+2→C
YX²÷C→Y
D+Y→D
WhileEnd
.5+De^(-X²÷2)÷√(2π◢1-Ans
```

示範 Example

```
TYPE  A=0, B=1, C=1.96   → GET z=1.96, 0.9750021049, 0.02499789515
TYPE  A=50, B=10, C=40   → GET z=-1, 0.1586552539, 0.8413447461
```

Paper 1: write the formula, substitution and reasoning. The program is a calculation/checking tool, not a replacement for method marks.